

Stopping LLMs from Going Rogue

A Control Barrier Approach to Text Generation

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Configuration Space & Geometry

Definition:

$$\mathcal{E}_W^n = \left\{ y \in \mathbb{R}^{n+1} \mid y^\top W y = 1 \right\}$$

$$\pi_W(x) = \frac{x}{\|x\|_W}$$

Numerical Example ($n = 1, k = 3, W = \mathbb{I}_2$):

$$y_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad y_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad y_3 = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

Insights:

- Tokens live on a defined geometric manifold[cite: 58].
- Normalization ensures tokens remain valid states within the ellipsoid[cite: 54].

Self-Attention Mechanism

Definition:

$$\alpha_{ij} = \frac{1}{Z_i} \exp(y_i^\top P y_j)$$

$$Z_i = \sqrt{n+1} \sum_{j=1}^k \exp(y_i^\top P y_j)$$

Numerical Example (Token 3 paying attention to 1, 2, 3):

$$Z_3 \approx 11.07 \implies \alpha_{31} = 0.090, \quad \alpha_{32} = 0.371, \quad \alpha_{33} = 0.246$$

Insights:

- Captures pairwise interactions between sequence tokens[cite: 62].
- Provides context-dependent weighting for polysemous words[cite: 62].

Attention Vector Field

Definition:

$$\tilde{f}_{sa,i}(y) = \sum_{j=1}^k \alpha_{ij} U y_j$$

Numerical Example (Weighted Sum for Token 3):

$$\tilde{f}_{sa,3}(y) = 0.090 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0.371 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0.246 \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix} = \begin{bmatrix} 0.264 \\ 0.545 \end{bmatrix}$$

Insights:

- Defines the direction of token movement based on context[cite: 71].
- Acts as a continuous interaction force pulling tokens together[cite: 71].

Feedforward Transformation

Definition:

$$\tilde{f}_{ff}(y) = W_2 \sigma(W_1 y + b_1) + b_2$$

Numerical Example (Applying ReLU and Maps):

$$\tilde{f}_{ff}(y_3) = 0.5 \begin{bmatrix} 1.414 \\ 0.707 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.807 \\ 0.454 \end{bmatrix}$$

Insights:

- Introduces a nonlinear transformation to the representation[cite: 71].
- Acts locally and independently on each individual token[cite: 71].

Discrete Transformer Dynamics

Definition:

$$y_i(l+1) = \pi_W \left(f_{sa,i} + \tau \tilde{f}_{ff}(f_{sa,i}) \right)$$

Numerical Example (One Full Update Step, $\tau = 0.1$):

$$\begin{aligned} y_3(l+1) &= \pi_W \left(\begin{bmatrix} 0.693 \\ 0.721 \end{bmatrix} + 0.1 \begin{bmatrix} 0.807 \\ 0.461 \end{bmatrix} \right) \\ &= \pi_W \left(\begin{bmatrix} 0.774 \\ 0.767 \end{bmatrix} \right) = \begin{bmatrix} 0.710 \\ 0.704 \end{bmatrix} \end{aligned}$$

Insights:

- Employs a residual structure common in modern architectures[cite: 67].
- The final projection enforces strict adherence to the geometry[cite: 71].

Definition:

$$\dot{y} = f_{sa}(t, y) + f_{ff}(t, y) + g(t, y_1) \cdot u$$

$$T_{y_i} \pi_W \cdot X = (\mathbb{I}_{n+1} - y_i y_i^\top) X$$

Insights:

- As $\tau \rightarrow 0$, finite differences converge into continuous tangent vectors[cite: 82].
- The discrete transformer becomes a fully continuous ODE system[cite: 87].
- Tangent projection naturally strips away non-manifold radial components[cite: 60].

Definition:

$$g(y_1) \cdot u = T_{y_1} \pi_W \cdot u$$

Numerical Example (Control Vector Projection):

$$u = \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} \implies T_{y_1} \pi_W \cdot u = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.2 \end{bmatrix}$$

Insights:

- The steering control input acts exclusively on the first token[cite: 75].
- Steering the prefix token controls the downstream sequence generation[cite: 79].
- Control Barrier Functions (CBFs) can bound this trajectory in a safe region[cite: 25].